Offroad Semantic Segmentation using DINOv2

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Abstract

This project focuses on semantic segmentation of off-road scenes using deep learning techniques.
Semantic segmentation is a computer vision task where each pixel of an image is assigned a class label.
In this project, a pre-trained DINOv2 Vision Transformer backbone is used along with a custom segmentation head
to classify different elements such as trees, rocks, sky, and ground.

The model was trained on a synthetic dataset with pixel-level annotations. Multiple experiments were conducted
using different numbers of epochs to observe performance variations. The final model achieved a validation IoU
of approximately 0.326 and a test IoU of approximately 0.233. The results demonstrate that the model is capable
of understanding complex off-road environments and generating meaningful segmentation outputs.

Introduction

In recent years, Artificial Intelligence and Machine Learning have played a crucial role in solving real-world problems.
One such problem is scene understanding, which is essential for applications like autonomous vehicles, robotics, and
environment monitoring.

Semantic segmentation provides a detailed understanding of an image by labeling each pixel instead of just identifying
objects. This project focuses on off-road environments, which are more complex than structured urban roads due to irregular
terrain, vegetation, and natural obstacles.

The aim of this project is to develop a model that can accurately segment different components of an off-road scene.

Problem Statement

The goal of this project is to build a semantic segmentation model that can classify each pixel in an off-road image
into predefined categories such as trees, bushes, rocks, and sky. The challenge lies in handling complex textures,
class imbalance, and small object detection.

Dataset Description

The dataset used in this project consists of synthetic desert environment images with pixel-level annotations.
It contains separate training, validation, and test sets. Each image has a corresponding segmentation mask where each pixel
represents a specific class.

Classes include:
Trees, Lush Bushes, Dry Grass, Dry Bushes, Ground Clutter, Flowers, Logs, Rocks, Landscape, Sky

Total Test Images: 10k+

The dataset helps in training the model to recognise various terrain elements in off-road conditions.

Methodology

The project follows a deep learning-based pipeline:

1. Data Preprocessing:
   Images are resized and normalised using standard ImageNet values.

2. Feature Extraction:
   A pre-trained DINOv2 Vision Transformer is used to extract high-level features from images.

3. Segmentation Head:
   A convolutional head processes the extracted features and generates pixel-wise predictions.

4. Training:
   The model is trained using CrossEntropy Loss and optimised using SGD.

5. Evaluation:
   The model is evaluated using metrics such as IoU, Dice Score, and Pixel Accuracy.

Model Architecture

The model consists of two main components:

1. Backbone (DINOv2):
   A pre-trained Vision Transformer that extracts feature embeddings from images.

2. Segmentation Head:
   A convolution-based module that converts embeddings into a segmentation map with 10 classes.

The architecture allows efficient feature extraction and accurate segmentation.

Training Details

Two experiments were conducted:

Experiment 1:
- Epochs: 10
- Validation IoU: ~0.324

Experiment 2:
- Epochs: 25
- Validation IoU: ~0.326

Observation:
Increasing epochs resulted in only slight improvement, indicating that the model reached convergence.

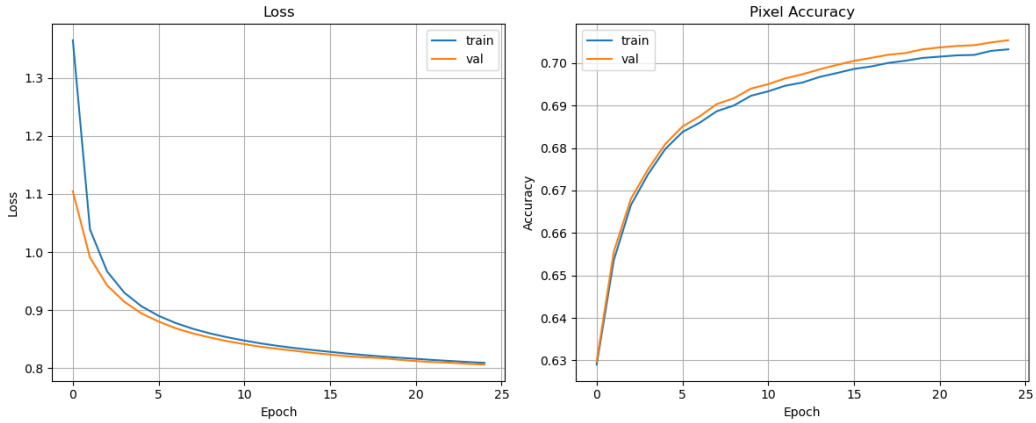
Results

Final Results:
- Validation IoU: ~0.326
- Test IoU: ~0.233
- Accuracy: ~70%

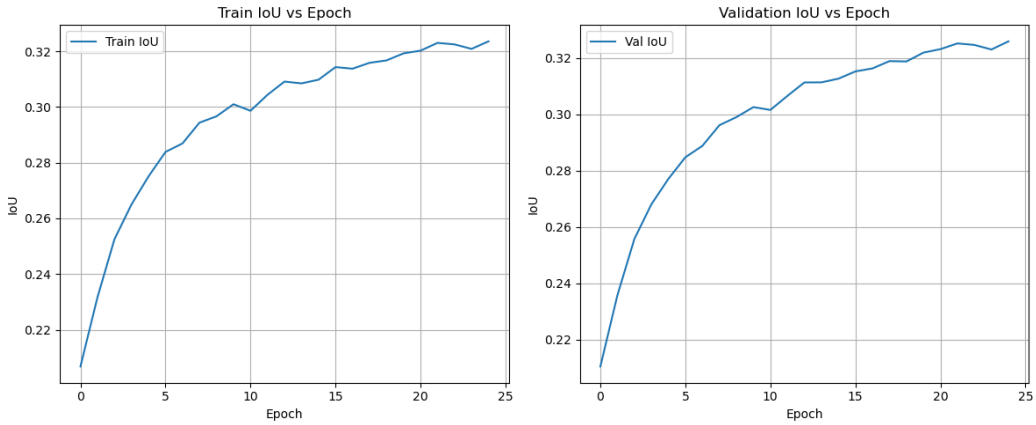
The model performs well on large regions such as sky and landscape, but struggles with smaller objects.

Results Visualisation

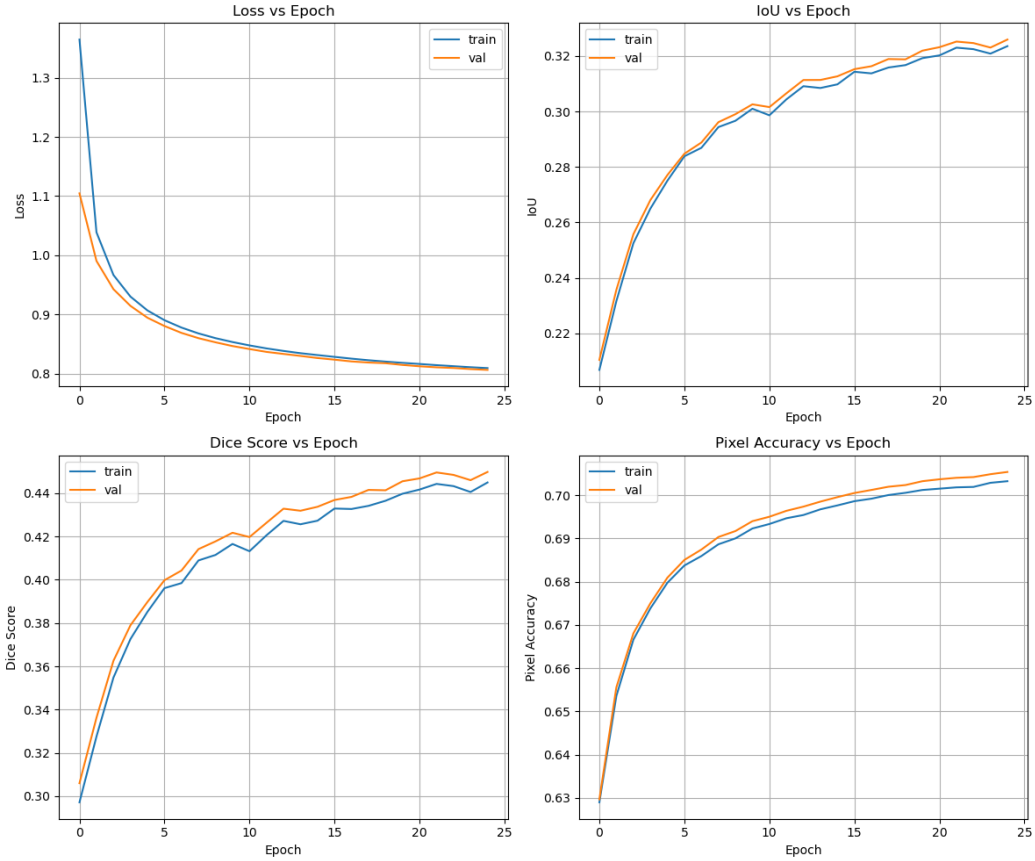
1. scripts/train_stats/training_curves.png



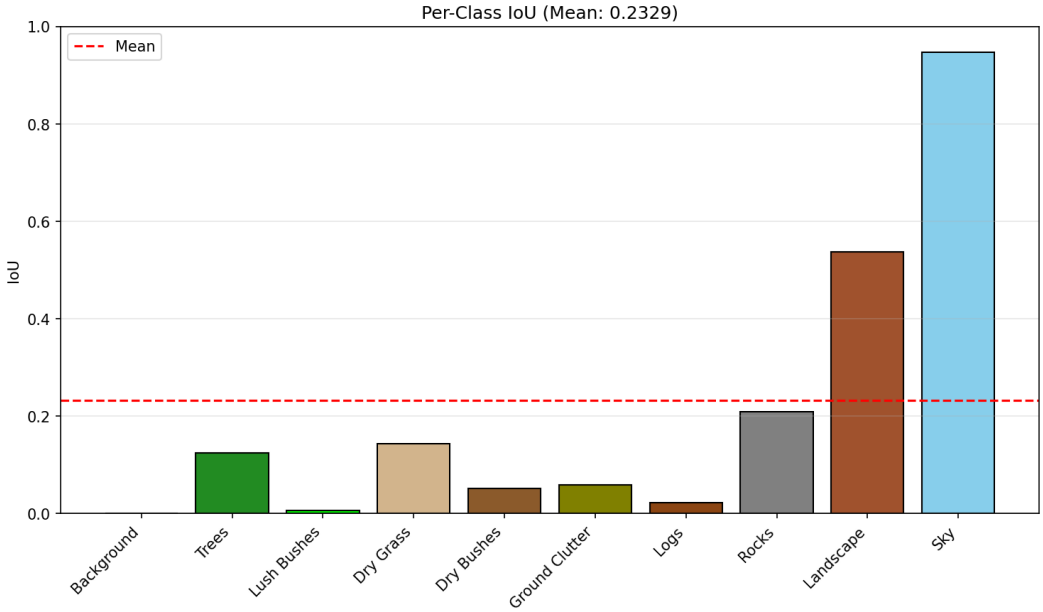
2. scripts/train_stats/iou_curves.png



3. scripts/train_stats/all_metrics_curves.png

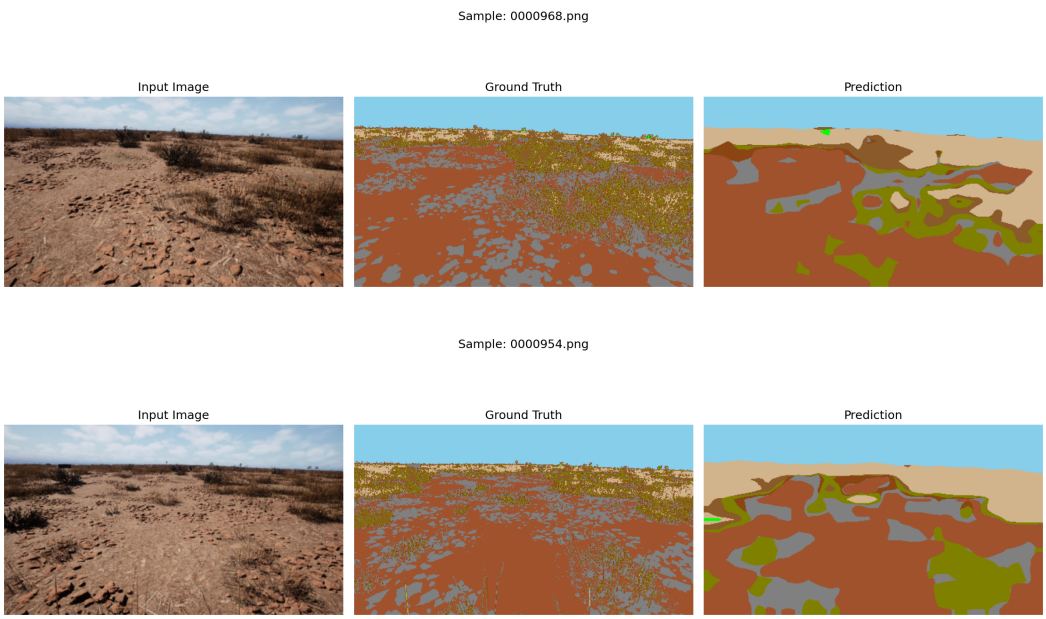


4. predictions/per_class_metrics.png



5. predictions/masks_color/ (sample image)

6. predictions/comparisons/ (sample comparison image)



Analysis

The model shows strong performance in detecting large continuous regions such as sky and landscape.
However, smaller objects like logs and bushes are difficult to detect due to fewer pixels and class imbalance.

Training curves indicate stable learning, and validation performance closely follows training performance,
suggesting good generalisation.

Challenges Faced

- Class imbalance in dataset
- Difficulty in detecting small objects
- Long training time due to CPU usage
- Limited hyper parameter tuning

Future Scope

- Use GPU for faster training
- Apply data augmentation
- Use advanced architectures like U-Net or DeepLabV3+
- Improve hyper parameter tuning

Conclusion

This project successfully demonstrates the use of deep learning for semantic segmentation of off-road scenes.
The model is capable of understanding complex environments and generating meaningful predictions.

Overall, the project highlights the practical application of AI and ML concepts in solving real-world problems.